

SALINA, KS, USA

WMO: 724586

Lat: 38.800N

Lon: 97.650W

Elev: 1269

StdP: 14.03

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 03919

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	4.6	9.3	-5.0	4.4	7.6	-0.6	5.6	11.8	32.1	34.7	28.2	36.8	9.6	330	0.602

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.5	101.7	73.9	98.6	74.0	95.3	73.5	77.5	92.5	76.4	91.2	75.2	89.7	15.7	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
(p)															
73.4	130.1	83.8	72.3	125.3	83.1	71.2	120.5	82.4	41.9	92.5	40.6	91.1	39.6	89.6	86.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
28.4	25.6	23.0	-2.7	106.6	5.2	2.7	-6.4	108.6	-9.5	110.1	-12.4	111.6	-16.1	113.5		
			-3.5	80.4	4.9	1.9	-7.1	81.7	-9.9	82.8	-12.7	83.9	-16.3	85.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
Temperatures, Degree-Days and Degree-Hours	DBAvg	56.8	31.4	35.4	45.8	55.4	66.2	77.1	81.8	79.6	71.2	58.0	44.9	33.4		
	DBStd	20.00	10.96	11.95	11.35	9.41	8.52	6.93	6.25	6.05	8.46	9.42	9.81	10.40		
	HDD50	2028	577	424	213	50	2	0	0	0	1	33	209	519		
	HDD65	4746	1041	828	600	309	93	4	0	0	35	254	603	979		
	CDD50	4508	1	16	81	213	504	812	984	916	636	283	57	5		
	CDD65	1752	0	0	4	22	130	366	520	451	220	38	1	0		
	CDH74	20430	0	3	67	290	1332	4237	6568	5181	2332	399	21	0		
	CDH80	10290	0	0	10	76	507	2137	3691	2710	1039	117	3	0		
	WSAvg	10.7	10.3	10.9	11.9	12.7	10.8	11.2	10.1	9.3	10.1	10.5	10.6	10.0		
	PrecAvg	31.7	0.7	1.0	2.4	3.0	4.2	4.5	3.8	3.3	2.8	2.4	1.4	0.9		
Precipitation	PrecMax	58.1	2.9	2.8	10.8	7.0	8.3	10.0	17.9	13.7	12.0	6.8	4.6	3.1		
	PrecMin	13.0	0.0	0.0	0.0	0.4	0.2	0.9	0.0	0.1	0.0	0.0	0.0	0.0		
	PrecStd	9.5	0.6	0.9	2.2	1.8	1.9	2.2	3.7	2.6	2.5	1.8	1.3	0.8		
	PrecStd	9.5	0.6	0.9	2.2	1.8	1.9	2.2	3.7	2.6	2.5	1.8	1.3	0.8		
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	65.5	73.0	80.6	87.6	95.1	102.3	106.3	104.1	98.9	89.4	76.9	65.1		
		MCWB	49.3	53.3	58.3	64.3	70.1	74.2	74.3	73.4	72.4	67.5	58.7	55.0		
	2%	DB	58.7	65.6	74.8	81.6	89.7	98.4	102.7	100.4	94.0	82.8	70.8	59.1		
		MCWB	45.6	51.1	58.3	63.2	69.8	73.9	74.3	73.6	71.4	64.5	56.0	48.4		
	5%	DB	53.0	59.5	69.9	76.8	85.6	94.9	99.4	96.8	90.2	78.4	65.6	54.3		
		MCWB	43.2	48.4	55.8	61.1	68.5	73.2	74.5	73.9	70.6	62.6	53.5	45.5		
	10%	DB	47.7	54.1	64.6	72.4	81.7	91.2	95.7	92.9	86.2	73.6	61.2	49.4		
		MCWB	40.3	45.0	53.3	59.1	67.4	72.2	74.3	73.3	69.0	61.2	52.0	42.0		
	0.4%	WB	51.4	57.7	65.5	68.8	74.8	78.3	79.4	79.0	76.2	71.5	61.7	57.5		
		MCDB	61.3	65.0	76.1	80.6	87.5	94.0	95.6	93.7	88.9	81.3	70.7	62.4		
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	46.9	53.3	61.4	66.1	72.7	76.5	77.6	77.2	74.2	68.5	59.0	50.5		
		MCDB	56.3	62.0	68.7	76.8	85.2	91.7	93.2	91.5	87.5	78.2	66.1	56.4		
	5%	WB	43.7	49.2	58.2	63.7	70.8	75.0	76.6	75.9	72.8	65.7	56.0	45.8		
		MCDB	52.8	59.7	67.3	74.1	82.2	89.9	91.9	90.0	85.8	74.7	64.5	53.5		
	10%	WB	40.3	44.9	54.0	60.7	68.8	73.7	75.4	74.6	71.2	62.6	52.4	42.2		
		MCDB	47.7	53.8	64.2	70.9	79.0	87.8	90.3	88.3	83.3	71.6	60.4	49.1		
Mean Daily Temperature Range	5% DB	MDBR	20.7	21.4	23.3	23.3	22.0	22.5	22.5	21.8	22.4	22.9	22.2	19.6		
		MCDBR	30.2	31.1	31.7	29.9	26.6	26.4	26.9	26.6	25.7	28.6	29.1	27.9		
		MCWBR	19.2	19.3	18.0	16.3	11.9	8.9	7.4	8.1	9.4	14.3	16.7	18.5		
	5% WB	MCDBR	28.9	28.8	25.2	25.8	23.1	23.9	23.0	22.6	22.1	22.2	24.4	25.7		
		MCWBR	19.1	18.9	16.4	16.0	11.9	9.9	8.3	9.0	9.7	13.2	15.7	18.2		
Clear-Sky Solar Irradiance	taub		0.284	0.294	0.328	0.362	0.393	0.409	0.419	0.415	0.377	0.340	0.295	0.279		
	taud		2.503	2.475	2.412	2.344	2.332	2.323	2.325	2.331	2.385	2.463	2.518	2.508		
	Ebn at Noon		287	298	296	290	281	276	272	271	275	276	280	280		
	Edh at Noon		24	29	34	39	40	40	40	39	35	29	24	23		
All-Sky Solar Radiation	RadAvg		762	1004	1334	1680	1924	2175	2135	1902	1565	1144	843	658		
	RadStd		68	112	98	99	145	114	129	103	91	120	71	62		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (21 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page